



1 · Data sources

Tufts Dental Database
1,000 panoramic · Universal

DENTEX (QED)
752 panoramic · FDI · 4 diseases

PR-Reports
121 cases · expert reports



2 · Ground-truth preparation

32-slot missing-teeth masks (Tufts)
abnormal-tooth + disease labels (DENTEX)
paired expert reports · readability checks



3 · Model Zoo (8 VLMs)

8 open-source VLMs

general · medical · dental · radiology
LLaVA-1.5 · LLaVA-Med · HuatuoGPT-V · DentVLM
Qwen2.5-VL · LLaVA-OneVision · MedGemma · LLaVA-Rad

× zero-shot · few-shot · chain-of-thought



6 · Scoring & metrics

Patient-level
Macro-F1 · accuracy · 95% bootstrap CI

Multi-label pathology
Hamming · Jaccard · Macro-4 F1

Report quality (RQ3)
BLEU-4 · ROUGE · BERTScore · KL/S



5 · Schema-constrained parser

Strict schema · explicit Universal ↔ FDI
empty / off-schema → all-negative → per-tooth



4 · Inference harness

Slurm on Deakin GPU host · fp16 (bf16 MedGemma)
per-model checkpointing → raw generations



7 · Hallucination audit

CHR = 1 – precision · per-image count
correct-condition / wrong-tooth vs. unsupported
plain vs. schema-reinforced ablation



8 · Outputs & synthesis

Summary tables · per-class & strategy figures
CHR & divergence plots · per-case JSON
→ research analysis & manuscript



Research questions

RQ1 · Diagnostic
general vs medical vs dental?

RQ2 · Prompting
zero / few / CoT effect?

RQ3 · Reporting
reports vs expert references?